

3590840

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BCA (Sem-V) – Network
Programming (504) DSE-II

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

Compulsory

1. Choose the correct alternative for each of the
following questions : 2×10=20

(i) Which protocol guarantees in-order delivery of data packets ?

- /a) TCP
- b) UDP
- c) SCTP
- d) All of these

(ii) Which protocol is faster but doesn't guarantee delivery ?

- a) TCP
- /b) UDP
- c) SCTP
- d) All of these

(iii) SCTP stands for :

- a) Secure connection Transmission Protocol
- /b) Stream Control Transmission Protocol

c) Simple Communication Transmission Protocol

d) Serialized Connection Transport Protocol

(iv) The server program tells its TCP that is ready to accept a connection. This process is called :

- /a) Active open
- b) Active close
- c) Passive close
- d) Passive open

(v) The port number is "ephemeral port number", if the source host is _____

- a) NTP
- b) Echo

c) Server

d) Client

(vi) HTTP uses the services of TCP on well known port number ____.

a) 88

b) 80

c) 404

d) 408

(vii) IMF stands for :

a) Internet Message Format

b) Inter Message Format

c) Internal Message Format

d) Intranct Message Format

(viii) The HTTP operates at which layer of the TCP/IP model ?

a) Network Access Layer

b) Internet Layer

c) Transport Layer

d) Application Layer

(ix) Socket options can be used for :

a) Define message content

b) Setting security feature

c) Specifying data transfer size

d) All of the above

(x) Socket acts like an endpoint for network communication similar to a :

a) Router

b) Firewall

c) Phone number

d) Cable

Section-B

Short Answer type Questions

Answer any four questions of the following :

5×4=20

2. Differentiate between TCP and UDP (any five).
3. What are some potential scenarios where SCTP might be more suitable choice than TCP and UDP.
4. Draw and explain UDP header.
5. Define any two email protocols.
6. Explain any five range of activities involved in the network management.
7. What is NVT ? Explain.

Section-C

Long Answer type Questions

Answer any two questions of the following :

15×2=30

8. Draw TCP header. List down all its fields.
Explain all fields.

9. WAP in java for client side socket.
10. What is I/O multiplexing ? Explain in details.
11. Explain the following :
 - (a) WWW
 - (b) HTTP
 - (c) IP

—————x—————